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PRODUCT DATASHEET · POWER SYSTEM TRAINING SIMULATOR

Karuma HPP Digital Twin

Generation – Transmission – Distribution simulator with integrated SCADA, protection relays, 3D training lab and instructor station



Real-time digital twin · Newton-Raphson load flow · Electromechanical dynamics · 65 protection functions · SCADA HMI · 3D training lab · Instructor & trainee workstations · Windows 10/11 · .exe

ASFAN

Software designed & developed by ASFAN

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Software version 1.0.0

Product overview

The **Karuma HPP Digital Twin** is a complete, real-time software replica of the Karuma hydro power plant training system. It reproduces the entire power chain — **prime mover, synchronous generator, step-up transformer, 220 kV transmission lines, substation switchgear & protection, distribution transformers with voltage regulation and distribution loads** — and presents it through a professional SCADA interface, protection relay panels, an interactive 3D training lab and a dedicated instructor station.

Trainees **monitor V, I, MW, MVAr and Hz**, **control breakers, loads and generation**, **analyse power flows** and **run training scenarios and fault conditions** exactly as they would on the physical training laboratory — safely, repeatably and without equipment wear. Every quantity is calculated by a physics-based network and machine model, not by animations or pre-recorded data.

FREQUENCY	TOTAL GENERATION	TOTAL LOAD	VOLTAGE (HV)
50.00 Hz	120 MW	118.5 MW	220 kV

Key features

- Physical power system training simulator: turbine trainer, generator, step-up transformer, transmission line simulator, substation, distribution transformer and loads
- SCADA system: Overview, Generation, Transmission, Distribution, Alarms, Reports and Trends pages
- Generator Control & AVR, Protection Relays, AGC & Network Control and SCADA interface panels
- Training room set-up: one instructor workstation and multiple trainee workstations over the LAN
- Real-time electromechanical simulation with automatic voltage, frequency and tap-changer control
- Short-circuit faults of any type at any location, cleared by realistic numerical protection

Typical applications

- Operator training for hydro power plants, transmission and distribution utilities
- University and technical-college power system laboratories
- Protection engineering courses (distance, differential, overcurrent, UFLS, auto-reclose)
- Black-start, islanding and grid-synchronisation drills
- Pre-commissioning familiarisation with the SCADA and switching procedures
- Trainee assessment with automatic scoring and printable reports

System architecture



SIMULATION CORE

Node-breaker topology processor · Newton-Raphson load flow per island · machine, turbine & control dynamics
Symmetrical-component fault analysis · protection relay engine · SCADA historian · alarms & SOE
Scenario assessment engine · classroom network server (instructor ⇄ trainee workstations)

SCADA – Power system overview

The SCADA overview shows the complete single-line diagram from the generator terminals to the customer loads, with live, colour-coded voltage levels, animated power flow, breaker states and the four key system indicators requested by the operator: **Frequency, Total Generation, Total Load and Voltage (HV)**. Any breaker can be operated directly from the diagram (with interlocking and synchro-check); clicking an equipment item opens its detailed page.

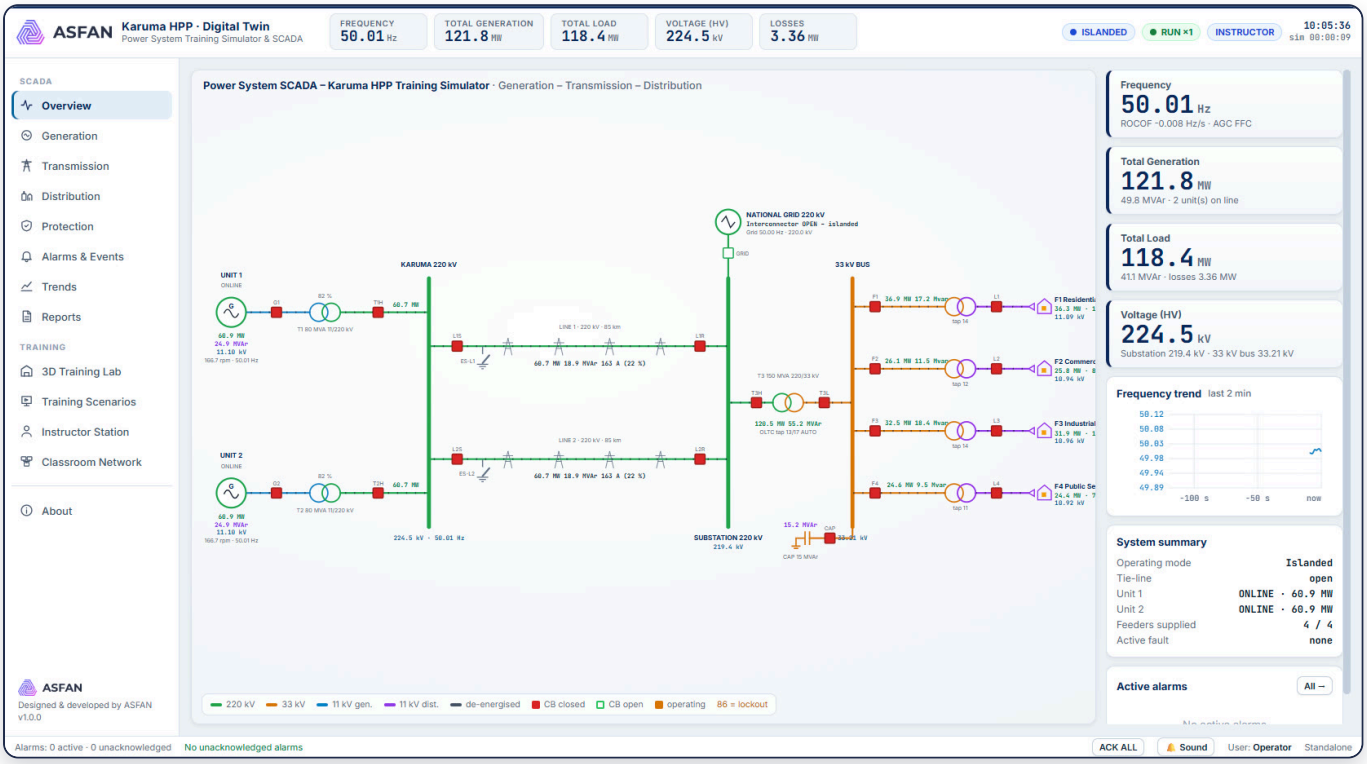


Figure 1 – SCADA overview: single-line diagram, key indicators, frequency trend, system summary and active alarms.

Simulated network

Section	Equipment represented
Generation	2 × 75 MVA, 11 kV, 0.85 pf hydro generators (63.75 MW each), Francis turbines with penstock, governors, static excitation & AVR, generator circuit breakers
Step-up	2 × 80 MVA, 11/231 kV Dyn11 step-up transformers with off-circuit taps, 220 kV breakers, Karuma 220 kV busbar
Transmission	220 kV double-circuit overhead line, 2 × 85 km, line breakers at both ends, line earthing switches, 2 500 MVA national-grid interconnector
Substation	Substation 220 kV busbar, 150 MVA 220/33 kV YNyn0 power transformer with on-load tap changer (±8 × 1.25 %), 33 kV busbar, 15 MVAr capacitor bank
Distribution	4 × 33 kV feeders (4–6 km), 4 distribution transformers 31.5–50 MVA 33/11 kV with OLTC & automatic voltage control, 11 kV incomers
Loads	Residential (36 MW), Commercial (26 MW), Industrial with motor load (32 MW), Public services & hospital (24.5 MW) – voltage- and frequency-dependent

Generation – Generator Control & AVR

Each unit has a full control faceplate reproducing the **Generator Control & AVR panel**: start / stop / emergency-stop sequences, speed and load set-points, governor mode (droop or isochronous), field breaker and AVR (auto / manual), synchroscope with manual and automatic synchronising, P–Q capability diagram and all turbine and generator measurements.

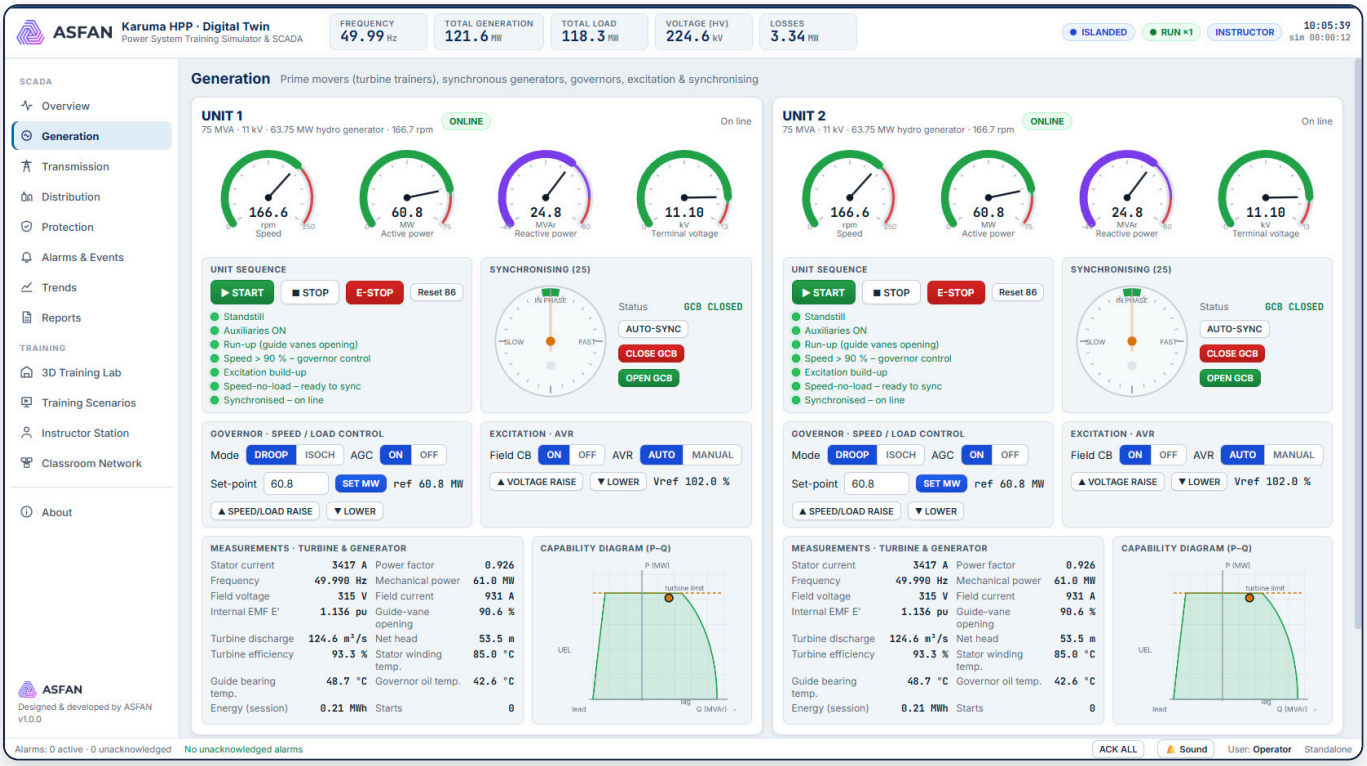


Figure 2 – Generation page with two unit faceplates, gauges, start sequence, synchroscope and capability diagram.

Synchronous generator & turbine (per unit)	
Rating	75 MVA, 11 kV, pf 0.85, 63.75 MW
Speed	166.7 rpm, 36 poles, 50 Hz
Reactances	Xd 1.05 · Xd' 0.30 · Xd'' 0.22 · X2 0.22 pu
Time constants	T'd0 6 s · H 4.0 s
Prime mover	Francis turbine, 54 m net head, water-hammer (Tw) model, guide-vane servo with rate limits
Measurements	V, I, MW, MVAr, pf, Hz, rpm, gate %, discharge, head, efficiency, field V/I, temperatures, energy

Controls	
Governor	PI speed/load controller, 4 % droop or isochronous, load ramp limiter, load-reference feed-forward
Excitation / AVR	Static exciter, PI AVR with reactive droop, V/Hz limiter, manual field control, field breaker
Synchronising	Synchroscope (Δθ, Δf, ΔV), synchro-check relay 25, auto-synchroniser
AGC & network control	OFF / flat-frequency / tie-line-bias modes, frequency bias, interchange schedule, ACE display
Sequences	Auxiliaries → run-up → speed control → excitation → speed-no-load → synchronised

Transmission & distribution



Figure 3 – Transmission: step-up transformers, 220 kV lines, grid synchronising, voltage profile, N-1 and short-circuit studies.

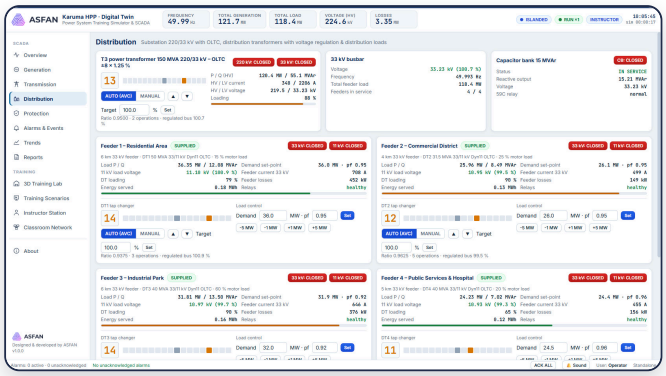


Figure 4 – Distribution: T3 OLTC, capacitor bank, four feeders with tap changers and load control.

Transmission functions

- Sending / receiving P, Q, current, losses and loading of each line and transformer
- Line parameters: Z1 = 0.06 + j0.41 Ω/km, Z0 = 0.30 + j1.15 Ω/km, shunt charging; rating 300 MVA
- Grid interconnector with synchroscope, synchro-check and grid frequency/voltage set by the instructor
- Busbar voltage magnitudes and angles; voltage profile from the generator to the load
- **N-1 contingency analysis** and **short-circuit level study** (3-phase and 1-phase, S″k, X/R) on demand, exportable to CSV

Distribution functions

- On-load tap changers on the 150 MVA substation transformer and all distribution transformers – AUTO (AVC) with dead-band and time delays, or MANUAL raise / lower
- 15 MVAr capacitor bank for reactive compensation
- Load demand and power-factor set-points, ±1 / ±5 MW load steps, 24-hour daily load profile
- Load models with voltage exponents and frequency sensitivity; motor load in the industrial park
- Energy served and energy-not-served accounting

Power flow & analysis engine

Topology	Node-breaker model with breakers and earthing switches; automatic bus and island detection
Load flow	Full Newton-Raphson AC load flow for every electrical island, distributed slack by unit inertia, recalculated every integration step
Dynamics	Swing equation per island, turbine/penstock, governor and excitation dynamics; time step 20 ms (5 ms during faults and switching)
Fault analysis	Symmetrical components (positive, negative, zero sequence) incl. transformer vector groups; 3PH, L-G, L-L, L-L-G faults with fault resistance at any bus or any % of a line

Protection relays

The protection system contains **65 numerical relay functions** (IEC 60255 / ANSI C37.2) evaluated in real time from simulated CT and VT signals. Relays start, time and trip breakers exactly like real devices, including breaker-failure back-up, lock-out (86) and auto-reclose. Each fault automatically creates a **disturbance record** with three-phase current and voltage waveforms and the trip sequence.

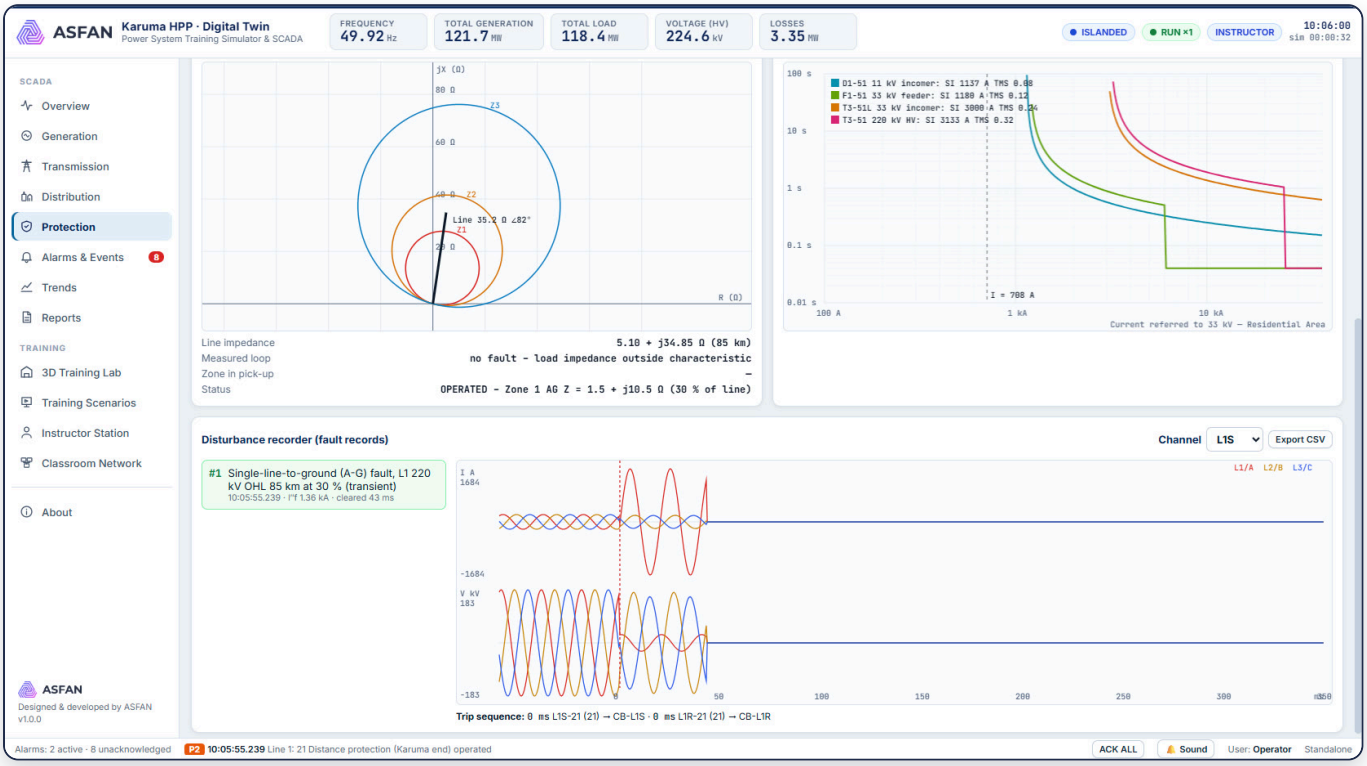


Figure 5 – Distance-protection R–X diagram, time–current coordination chart and disturbance recorder after a single-phase line fault cleared in 45 ms.

Protected zone	Functions (ANSI)
Generators	**87G** differential · **64G** stator earth fault · **51V** voltage-restrained O/C · **46** negative sequence · **32** reverse power · **40** loss of excitation · **81U/81O** · **59/27**
Transformers	**87T** differential · **50/51** phase O/C (IEC SI/VI/EI curves) · **51N** earth fault · 33 kV and 11 kV incomer O/C
Busbars	**87B** busbar differential (Karuma 220 kV, substation 220 kV, 33 kV)
220 kV lines	**21** distance (mho, Z1 80 %, Z2 120 %, Z3 220 %, POTT, six loops with k0) · **67N** directional E/F · **79** auto-reclose with lock-out
Grid tie	**67** directional O/C · **81R/78** islanding (ROCOF) · **25** synchro-check
Distribution	Feeder **50/51**, **50N/51N** · DT **87T** · 3-stage under-frequency load shedding **81U** · motor **27** · capacitor **59C**
All breakers	**50BF** breaker-failure back-up tripping · **86** lock-out with reset · switching interlocks · earthing-switch interlocks

SCADA – Alarms, trends & reports

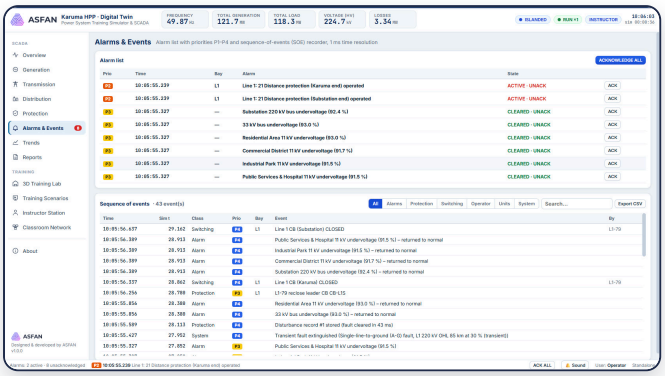


Figure 6 – Alarm list (P1–P4) and sequence-of-events recorder.

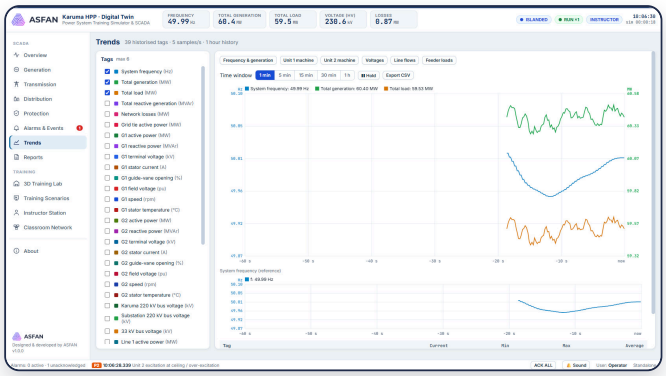


Figure 7 – Trends: 39 historised tags, up to 6 pens, 1 min – 1 h windows.

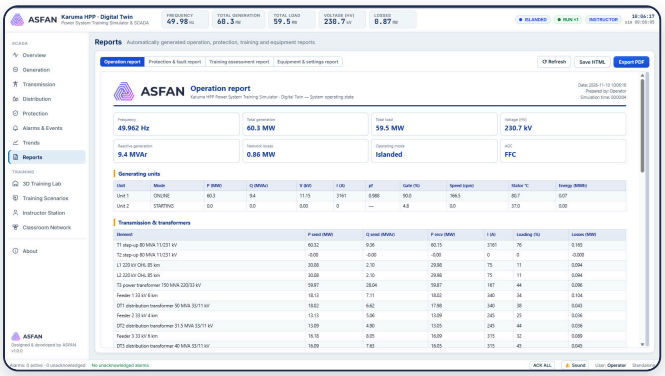


Figure 8 – Automatic reports with PDF export.

Alarms & events

- Four alarm priorities, audible alarm, acknowledge single / all, alarm ticker
- Sequence of events with 1 ms resolution: switching, protection, operator, unit and system events; filter, search and CSV export

Trends

- 1 hour history at 5 samples per second, hold / resume, statistics (min / max / average), CSV export

Reports

- Operation report · Protection & fault report · Training assessment report · Equipment & settings report – exported as PDF or HTML

Training lab, scenarios & instructor station



Figure 9 – Interactive 3D training lab with live equipment states; click any panel to open its SCADA page.

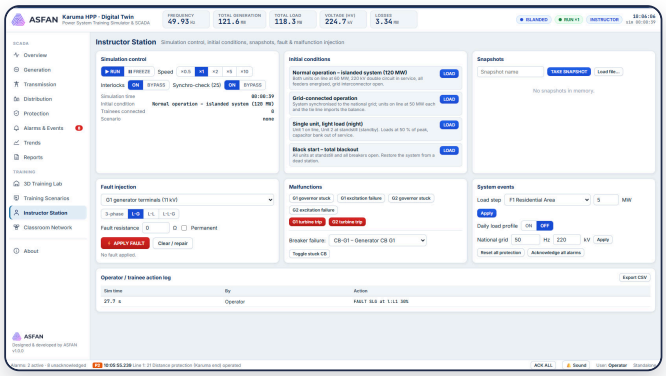


Figure 10 – Instructor station: run / freeze / speed, initial conditions, snapshots, faults and malfunctions.

Training scenarios with automatic assessment

Scenario	Level	Duration
Unit start-up & manual synchronisation	BASIC	12 min
Frequency control after a load increase	BASIC	8 min
220 kV line fault – isolation & restoration	INTERMEDIATE	15 min
Voltage regulation with OLTC, capacitor bank & AVR	INTERMEDIATE	10 min
Synchronisation to the national grid & tie-line control	INTERMEDIATE	10 min
Generator trip, under-frequency load shedding & recovery	ADVANCED	15 min
Black start & system restoration	ADVANCED	25 min
Feeder fault with breaker failure (50BF)	ADVANCED	12 min

Instructor station

- 4 initial conditions: islanded, grid-connected, single unit / light load, total blackout
- Snapshots save / restore (memory and file)
- Fault injection: 3PH, L-G, L-L, L-L-G, fault resistance, transient or permanent, any bus or line position
- Malfunctions: governor stuck, excitation failure, turbine trip, stuck breaker; interlock and synchro-check bypass
- Operator / trainee action log with CSV export

Classroom network

- Instructor workstation hosts the simulation; trainee workstations join over the training-room LAN
- All trainees see the same live plant and operate it from their own SCADA screens
- Instructor-only commands are protected; each trainee action is logged by name
- Scores, objectives, penalties and quiz answers stored in the training assessment report

Technical specifications

Software	
Product	Karuma HPP Digital Twin – Power System Training Simulator & SCADA, version 1.0.0
Delivery	Windows installer (Setup .exe) and portable .exe – no additional runtime required
Operating modes	Standalone · Instructor workstation (host) · Trainee workstation (client)
Simulation	Real time with ×0.25 – ×10 speed and freeze; 20 ms step (5 ms during transients)
Data	39 historised tags (1 h, 5 Hz), 5 000-event SOE buffer, 20 disturbance records
Exports	PDF and HTML reports; CSV for events, trends, studies, disturbance records and action log; JSON snapshots
Network	TCP/IP LAN, HTTP + Server-Sent Events, default port 47820
Language / units	English user interface, SI units, IEC / ANSI device numbering

Recommended workstation	
Operating system	Windows 10 or Windows 11, 64-bit
Processor / memory	4-core CPU, 2.5 GHz or faster · 8 GB RAM (16 GB recommended for the instructor workstation)
Graphics	DirectX 11 / WebGL 2 capable GPU for the 3D training lab
Display	1920 × 1080 or higher (dual monitors recommended for the instructor)
Disk	500 MB free space

Scope of supply	
Software	Karuma HPP Digital Twin installer and portable executable
Documentation	This product datasheet (included in the installation)
Services (optional)	Installation and commissioning, instructor training, additional customer-specific scenarios, model adaptation to site data

ASFAN

Software developer: ASFAN

The Karuma HPP Digital Twin software — simulation models, SCADA, protection, 3D training lab and instructor tools — is **designed and developed by ASFAN**.

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